

BUSINESS REQUIREMENTS DOCUMENT

E-Commerce UI Test Automation Platform

Project: Playwright + TypeScript E-Commerce Test Automation Framework

Application under test: AutomationExercise

GitHub Repository: paritk7210-tester/TypeScript

Document Version: 1.0

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Status: Portfolio Project

1. Executive Summary

This BRD defines the business and functional requirements for an E-Commerce UI Test Automation Framework built with Playwright and TypeScript. The objective is to provide repeatable, maintainable and traceable validation of critical customer journeys such as registration, login, product discovery, cart management, checkout, payment, contact, subscription and related flows.

The project demonstrates a BA-to-testing lifecycle: understanding business journeys, translating them into testable requirements and acceptance criteria, identifying risks and dependencies, prioritizing scope, and validating the implemented application through automated end-to-end tests.

2. Business Background / Problem Statement

Manual regression testing of e-commerce journeys can be time-consuming, inconsistent and difficult to repeat across browsers. A reusable automation framework is required to improve regression coverage, reduce repetitive execution effort, provide faster feedback and create auditable test evidence.

3. Business Objectives

ID	Objective
OBJ-01	Automate critical e-commerce customer journeys end-to-end.
OBJ-02	Reduce repetitive manual regression effort through reusable automation.
OBJ-03	Support cross-browser validation across Chromium, Edge and Firefox.
OBJ-04	Use maintainable Page Object Model design and reusable fixtures/utilities.
OBJ-05	Provide test evidence through HTML reporting, screenshots, videos and traces.
OBJ-06	Maintain centralized, JSON-based test data for scalable data-driven testing.

4. Project Scope

In Scope

- User registration and login
- Product search and product details
- Cart and quantity management
- Checkout and shipping flow
- Payment-page validation
- Contact form and file upload

- Subscription flow
- Test Cases page and related navigation
- Cross-browser execution
- JSON test data, fixtures, reusable page classes and common utilities
- HTML reports, screenshots, videos and Playwright Trace Viewer evidence

Out of Scope

- Production payment processing or real financial transactions
- Backend/service ownership or production infrastructure changes
- Native mobile application testing
- Business ownership of the AutomationExercise application
- Performance/load testing unless separately implemented

5. Stakeholders

Stakeholder	Responsibility
Business Analyst / Product perspective	Elicit business journeys, define requirements, acceptance criteria and priorities.
Test Analyst / QA	Design and execute functional, regression and automated test coverage.
Automation Engineer	Develop and maintain Playwright/TypeScript framework components.
Developer / Engineering	Resolve application defects and support testability.
Business / Product Owner	Validate expected customer journeys and business outcomes.

6. Business Requirements

ID	Business Requirement	Priority
BR-01	The system shall allow a customer to register and authenticate successfully.	Must Have
BR-02	The system shall allow users to search/browse products and view product details.	Must Have
BR-03	The system shall allow users to add products to the cart and manage cart contents.	Must Have
BR-04	The system shall allow a user to proceed through the checkout journey.	Must Have
BR-05	The system shall validate the payment-page journey after checkout information is submitted.	Must Have
BR-06	The framework shall automate regression scenarios using reusable page objects.	Must Have
BR-07	The framework shall support execution on Chromium, Edge and Firefox.	Should Have
BR-08	The framework shall use centralized JSON test data for applicable scenarios.	Should Have
BR-09	The framework shall generate HTML execution evidence and support screenshots, video and traces.	Should Have
BR-10	The framework shall support maintainable fixtures and common utility functions.	Should Have

7. Functional Requirements & Acceptance Criteria

ID	Feature	Acceptance Criteria
FR-01	Registration	Valid registration details create a user account; invalid/duplicate inputs are handled with appropriate validation.
FR-02	Login	Valid credentials authenticate the user; invalid credentials are rejected.
FR-03	Product Search	User can navigate to products, search/filter where supported and open product details.
FR-04	Cart	User can add products, view cart and validate selected items/quantities.

ID	Feature	Acceptance Criteria
FR-05	Checkout	User can enter/confirm checkout information and progress through the expected flow.
FR-06	Payment	Payment-page navigation and expected page state are validated without performing a real financial transaction.
FR-07	Contact	User can submit the contact form and validate the expected confirmation behavior.
FR-08	Subscription	User can submit subscription details and validate the expected response.
FR-09	Cross-browser	Applicable automated tests execute successfully against Chromium, Edge and Firefox.
FR-10	Reporting	Execution produces an HTML report and supporting evidence for failed/diagnostic scenarios.

8. Non-Functional Requirements

Category	Requirement
Maintainability	Use Page Object Model, reusable fixtures, common actions and separated test data.
Reliability	Tests should use stable locators and appropriate synchronization to reduce flaky execution.
Portability	Framework should support configured browsers including Chromium, Edge and Firefox.
Traceability	Test names and data should clearly map to the customer journey being validated.
Reporting	Execution should provide HTML reporting plus screenshots/videos/traces where configured.
Scalability	Framework structure should allow additional pages and scenarios without duplicating common logic.

9. Assumptions and Constraints

- AutomationExercise remains accessible and its UI behavior is sufficiently stable for automated validation.
- Test credentials and data used by the framework are non-production and appropriate for test execution.
- Third-party services or application changes may cause test failures unrelated to framework defects.
- Real payment processing is not performed; payment validation is limited to the available application test flow.

10. Risks and Mitigation

Risk	Impact	Mitigation
UI locator changes	High	Prefer stable locators; centralize selectors within page objects.
Application availability issues	Medium	Separate environment/application failures from automation defects and retain evidence.
Flaky synchronization	High	Use Playwright waits/assertions appropriately rather than arbitrary delays.
Test-data conflicts	Medium	Centralize and manage JSON test data; use unique data where required.
Cross-browser differences	Medium	Run the regression suite against configured browsers and investigate browser-specific failures.

11. Traceability / BA Perspective

The framework can be presented as a BA-led requirements-to-validation workflow: Business Journey → Business Requirement → Functional Requirement → Acceptance Criteria → Test Scenario → Automated Test → Execution Evidence → Defect / Retest → Regression Sign-off. This demonstrates that automation is not only a coding exercise but a mechanism for validating defined business outcomes.

12. Deliverables

- Business Requirements Document (BRD)
- Functional requirements and acceptance criteria

- Playwright + TypeScript automation framework
- Page Object Model components
- Fixtures, utilities and centralized JSON test data
- Automated end-to-end regression scenarios
- Cross-browser execution configuration
- HTML test reports and diagnostic evidence
- GitHub repository: [paritk7210-tester/TypeScript](#)

13. Success Criteria

- Critical customer journeys are covered by automated end-to-end tests.
- Automation executes consistently across the configured browser matrix.
- Test failures provide actionable evidence through reports, screenshots, videos or traces where enabled.
- New scenarios can be added using existing page objects, fixtures and utilities without unnecessary duplication.
- Requirements and acceptance criteria can be traced to corresponding automated scenarios.

14. Approval

This portfolio BRD is prepared to demonstrate BA analysis and requirements documentation for the Playwright testing project. In a production delivery, requirements would be baselined after stakeholder review and approval.